O u t[] =

{{A, 1}, {π+, 2}, {π-, 3}}  e p2

μ1 -  e p3

μ1

{{A, 1}, {π+, 2}, {W-, 3}} - e2 Vud ημ1,μ3 fπ

2 sw

{{A, 1}, {π-, 2}, {W+, 3}} e2 Vud ημ1,μ3 fπ

2 sw

Δ0, 1, Δ0, 2, {Z, 3}  e γs1,s2

μ3 ημ1,μ2

6 cw sw

+  cw e γs1,s2

μ3 ημ1,μ2

12 sw

+  e sw γs1,s2

μ3 ημ1,μ2

12 cw

+  AD e γs1,s2

μ2 ημ1,μ3

6 cw sw

+  AD cw e γs1,s2

μ2 ημ1,μ3

12 sw

+  AD e sw γs1,s2

μ2 ημ1,μ3

12 cw

+

 AD e γs1,s2

μ1 ημ2,μ3

6 cw sw

+  AD cw e γs1,s2

μ1 ημ2,μ3

12 sw

+  AD e sw γs1,s2

μ1 ημ2,μ3

12 cw

+  e γμ1.γμ3.γμ2s1,s2
12 cw sw

+  AD e γμ1.γμ3.γμ2s1,s2
6 cw sw

+  AD2 e γμ1.γμ3.γμ2s1,s2
4 cw sw

+

 cw e γμ1.γμ3.γμ2s1,s2

24 sw

+  AD cw e γμ1.γμ3.γμ2s1,s2
12 sw

+  AD2 cw e γμ1.γμ3.γμ2s1,s2
8 sw

+  e sw γμ1.γμ3.γμ2s1,s2
24 cw

+  AD e sw γμ1.γμ3.γμ2s1,s2
12 cw

+  AD2 e sw γμ1.γμ3.γμ2s1,s2
8 cw

Δ0, 1, {Δ-, 2}, {W+, 3} -

 3
2 e Vud γs1,s2

μ3 ημ1,μ2

2 sw

-

 3
2 AD e Vud γs1,s2

μ2 ημ1,μ3

2 sw

-

 3
2 AD e Vud γs1,s2

μ1 ημ2,μ3

2 sw

-

 3
2 e Vud γμ1.γμ3.γμ2s1,s2
4 sw

-

 3
2 AD e Vud γμ1.γμ3.γμ2s1,s2
2 sw

-

3  3
2 AD2 e Vud γμ1.γμ3.γμ2s1,s2
4 sw

Δ0, 1, {Δ+, 2}, {W-, 3} -  e Vud γs1,s2

μ3 ημ1,μ2

2 sw

-  AD e Vud γs1,s2

μ2 ημ1,μ3

2 sw

-  AD e Vud γs1,s2

μ1 ημ2,μ3

2 sw

-  e Vud γμ1.γμ3.γμ2s1,s2
2 2 sw

-  AD e Vud γμ1.γμ3.γμ2s1,s2
2 sw

- 3  AD2 e Vud γμ1.γμ3.γμ2s1,s2
2 2 sw

Δ0, 1, {n, 2}, {A, 3} - 2

3 G2 Kappa p1

μ3 p3

μ1 γs1,s2

5 + G1 Kappa ϵμ1,μ3,α1,β1 p1

α1 p3

β1 δs1,s2
6 + G1 Kappa ϵμ1,μ3,α1,γ1 p1

α1 p3

γ1 δs1,s2
6 +  2

3 G2 Kappa γs1,s2

5 ημ1,μ3 p1.p3

Δ0, 1, {n, 2}, π0, 3 -

 2
3 hD ϵμ1,α1,β1,γ1 p1

β1 p3

α1 γγ1.γ5

s1,s2
MDelta fπ

Δ0, 1, {n, 2}, {Z, 3} - e hD ϵμ1,μ3,α1,β1 p1

α1 γβ1.γ5

s1,s2
6 cw MDelta sw

Δ0, 1, {p, 2}, {π-, 3} -  hD ϵμ1,α1,β1,γ1 p1

β1 p3

α1 γγ1.γ5

s1,s2
3 MDelta fπ

Δ0, 1, {p, 2}, {W-, 3} - e hD Vud ϵμ1,μ3,α1,β1 p1

α1 γβ1.γ5

s1,s2
2 3 MDelta sw

{Δ-, 1}, Δ0, 2, {W-, 3} -

 3
2 e Vud γs1,s2

μ3 ημ1,μ2

2 sw

-

 3
2 AD e Vud γs1,s2

μ2 ημ1,μ3

2 sw

-

 3
2 AD e Vud γs1,s2

μ1 ημ2,μ3

2 sw

-

 3
2 e Vud γμ1.γμ3.γμ2s1,s2
4 sw

-

 3
2 AD e Vud γμ1.γμ3.γμ2s1,s2
2 sw

-

3  3
2 AD2 e Vud γμ1.γμ3.γμ2s1,s2
4 sw

{{Δ-, 1}, {Δ-, 2}, {A, 3}}  e γs1,s2

μ3 ημ1,μ2 +  AD e γs1,s2

μ2 ημ1,μ3 +  AD e γs1,s2

μ1 ημ2,μ3 + 1

2  e γμ1.γμ3.γμ2s1,s2 +  AD e γμ1.γμ3.γμ2s1,s2 + 3

2  AD2 e γμ1.γμ3.γμ2s1,s2

{{Δ-, 1}, {Δ-, 2}, {Z, 3}}

 e γs1,s2

μ3 ημ1,μ2

2 cw sw

+  cw e γs1,s2

μ3 ημ1,μ2

4 sw

- 3  e sw γs1,s2

μ3 ημ1,μ2

4 cw

+  AD e γs1,s2

μ2 ημ1,μ3
2 cw sw

+  AD cw e γs1,s2

μ2 ημ1,μ3
4 sw

- 3  AD e sw γs1,s2

μ2 ημ1,μ3
4 cw

+

 AD e γs1,s2

μ1 ημ2,μ3

2 cw sw

+  AD cw e γs1,s2

μ1 ημ2,μ3

4 sw

- 3  AD e sw γs1,s2

μ1 ημ2,μ3

4 cw

+  e γμ1.γμ3.γμ2s1,s2
4 cw sw

+  AD e γμ1.γμ3.γμ2s1,s2
2 cw sw

+ 3  AD2 e γμ1.γμ3.γμ2s1,s2
4 cw sw

+

 cw e γμ1.γμ3.γμ2s1,s2

8 sw

+  AD cw e γμ1.γμ3.γμ2s1,s2
4 sw

+ 3  AD2 cw e γμ1.γμ3.γμ2s1,s2
8 sw

- 3  e sw γμ1.γμ3.γμ2s1,s2
8 cw

- 3  AD e sw γμ1.γμ3.γμ2s1,s2
4 cw

- 9  AD2 e sw γμ1.γμ3.γμ2s1,s2
8 cw

{{Δ-, 1}, {n, 2}, {π-, 3}} -  hD ϵμ1,α1,β1,γ1 p1

β1 p3

α1 γγ1.γ5

s1,s2
MDelta fπ

{{Δ-, 1}, {n, 2}, {W-, 3}} - e hD Vud ϵμ1,μ3,α1,β1 p1

α1 γβ1.γ5

s1,s2
2 MDelta sw

Δ+, 1, Δ0, 2, {W+, 3} -  e Vud γs1,s2

μ3 ημ1,μ2

2 sw

-  AD e Vud γs1,s2

μ2 ημ1,μ3
2 sw

-  AD e Vud γs1,s2

μ1 ημ2,μ3
2 sw

-  e Vud γμ1.γμ3.γμ2s1,s2
2 2 sw

-  AD e Vud γμ1.γμ3.γμ2s1,s2
2 sw

- 3  AD2 e Vud γμ1.γμ3.γμ2s1,s2
2 2 sw

Δ+, 1, {Δ+, 2}, {A, 3} - e γs1,s2

μ3 ημ1,μ2 -  AD e γs1,s2

μ2 ημ1,μ3 -  AD e γs1,s2

μ1 ημ2,μ3 - 1

2  e γμ1.γμ3.γμ2s1,s2 -  AD e γμ1.γμ3.γμ2s1,s2 - 3

2  AD2 e γμ1.γμ3.γμ2s1,s2

Δ+, 1, {Δ+, 2}, {Z, 3} -  e γs1,s2

μ3 ημ1,μ2

6 cw sw

-  cw e γs1,s2

μ3 ημ1,μ2

12 sw

+ 11  e sw γs1,s2

μ3 ημ1,μ2

12 cw

-  AD e γs1,s2

μ2 ημ1,μ3

6 cw sw

-  AD cw e γs1,s2

μ2 ημ1,μ3

12 sw

+ 11  AD e sw γs1,s2

μ2 ημ1,μ3

12 cw

-

 AD e γs1,s2

μ1 ημ2,μ3
6 cw sw

-  AD cw e γs1,s2

μ1 ημ2,μ3
12 sw

+ 11  AD e sw γs1,s2

μ1 ημ2,μ3
12 cw

-  e γμ1.γμ3.γμ2s1,s2
12 cw sw

-  AD e γμ1.γμ3.γμ2s1,s2
6 cw sw

-  AD2 e γμ1.γμ3.γμ2s1,s2
4 cw sw

-

 cw e γμ1.γμ3.γμ2s1,s2

24 sw

-  AD cw e γμ1.γμ3.γμ2s1,s2
12 sw

-  AD2 cw e γμ1.γμ3.γμ2s1,s2
8 sw

+ 11  e sw γμ1.γμ3.γμ2s1,s2
24 cw

+ 11  AD e sw γμ1.γμ3.γμ2s1,s2
12 cw

+ 11  AD2 e sw γμ1.γμ3.γμ2s1,s2
8 cw

Δ+, 1, {Δ++, 2}, {W-, 3} -

 3
2 e Vud γs1,s2

μ3 ημ1,μ2

2 sw

-

 3
2 AD e Vud γs1,s2

μ2 ημ1,μ3

2 sw

-

 3
2 AD e Vud γs1,s2

μ1 ημ2,μ3

2 sw

-

 3
2 e Vud γμ1.γμ3.γμ2s1,s2
4 sw

-

 3
2 AD e Vud γμ1.γμ3.γμ2s1,s2
2 sw

-

3  3
2 AD2 e Vud γμ1.γμ3.γμ2s1,s2
4 sw

Δ+, 1, {n, 2}, {π+, 3}  hD ϵμ1,α1,β1,γ1 p1

β1 p3

α1 γγ1.γ5

s1,s2
3 MDelta fπ

Δ+, 1, {n, 2}, {W+, 3} e hD Vud ϵμ1,μ3,α1,β1 p1

α1 γβ1.γ5

s1,s2
2 3 MDelta sw

Δ+, 1, {p, 2}, {A, 3} - 2

3 G2 Kappa p1

μ3 p3

μ1 γs1,s2

5 + G1 Kappa ϵμ1,μ3,α1,β1 p1

α1 p3

β1 δs1,s2
6 + G1 Kappa ϵμ1,μ3,α1,γ1 p1

α1 p3

γ1 δs1,s2
6 +  2

3 G2 Kappa γs1,s2

5 ημ1,μ3 p1.p3

Δ+, 1, {p, 2}, π0, 3 -

 2
3 hD ϵμ1,α1,β1,γ1 p1

β1 p3

α1 γγ1.γ5

s1,s2
MDelta fπ

Δ+, 1, {p, 2}, {Z, 3} - e hD ϵμ1,μ3,α1,β1 p1

α1 γβ1.γ5

s1,s2
6 cw MDelta sw

Δ++, 1, {Δ+, 2}, {W+, 3} -

 3
2 e Vud γs1,s2

μ3 ημ1,μ2

2 sw

-

 3
2 AD e Vud γs1,s2

μ2 ημ1,μ3

2 sw

-

 3
2 AD e Vud γs1,s2

μ1 ημ2,μ3

2 sw

-

 3
2 e Vud γμ1.γμ3.γμ2s1,s2
4 sw

-

 3
2 AD e Vud γμ1.γμ3.γμ2s1,s2
2 sw

-

3  3
2 AD2 e Vud γμ1.γμ3.γμ2s1,s2
4 sw

Δ++, 1, {Δ++, 2}, {A, 3} -2  e γs1,s2

μ3 ημ1,μ2 - 2  AD e γs1,s2

μ2 ημ1,μ3 - 2  AD e γs1,s2

μ1 ημ2,μ3 -  e γμ1.γμ3.γμ2s1,s2 - 2  AD e γμ1.γμ3.γμ2s1,s2 - 3  AD2 e γμ1.γμ3.γμ2s1,s2

Δ++, 1, {Δ++, 2}, {Z, 3} -  e γs1,s2

μ3 ημ1,μ2

2 cw sw

-  cw e γs1,s2

μ3 ημ1,μ2

4 sw

+ 7  e sw γs1,s2

μ3 ημ1,μ2

4 cw

-  AD e γs1,s2

μ2 ημ1,μ3

2 cw sw

-  AD cw e γs1,s2

μ2 ημ1,μ3

4 sw

+ 7  AD e sw γs1,s2

μ2 ημ1,μ3

4 cw

-

 AD e γs1,s2

μ1 ημ2,μ3
2 cw sw

-  AD cw e γs1,s2

μ1 ημ2,μ3
4 sw

+ 7  AD e sw γs1,s2

μ1 ημ2,μ3
4 cw

-  e γμ1.γμ3.γμ2s1,s2
4 cw sw

-  AD e γμ1.γμ3.γμ2s1,s2
2 cw sw

- 3  AD2 e γμ1.γμ3.γμ2s1,s2
4 cw sw

-

 cw e γμ1.γμ3.γμ2s1,s2

8 sw

-  AD cw e γμ1.γμ3.γμ2s1,s2
4 sw

- 3  AD2 cw e γμ1.γμ3.γμ2s1,s2
8 sw

+ 7  e sw γμ1.γμ3.γμ2s1,s2
8 cw

+ 7  AD e sw γμ1.γμ3.γμ2s1,s2
4 cw

+ 21  AD2 e sw γμ1.γμ3.γμ2s1,s2
8 cw

Δ++, 1, {p, 2}, {π+, 3}  hD ϵμ1,α1,β1,γ1 p1

β1 p3

α1 γγ1.γ5

s1,s2
MDelta fπ

Δ++, 1, {p, 2}, {W+, 3} e hD Vud ϵμ1,μ3,α1,β1 p1

α1 γβ1.γ5

s1,s2
2 MDelta sw

{n, 1}, Δ0, 2, {A, 3}  2

3 G2 Kappa p2

μ3 p3

μ2 γs1,s2

5 - G1 Kappa ϵμ2,μ3,α1,β1 p2

α1 p3

β1 δs2,s1
6 - G1 Kappa ϵμ2,μ3,α1,γ1 p2

α1 p3

γ1 δs2,s1
6 -  2

3 G2 Kappa γs1,s2

5 ημ2,μ3 p2.p3

{n, 1}, Δ0, 2, π0, 3 -

 2
3 hD ϵμ2,α1,β1,γ1 p2

β1 p3

α1 γγ1.γ5

s1,s2

MDelta fπ

{n, 1}, Δ0, 2, {Z, 3} - e hD ϵμ2,μ3,α1,β1 p2

α1 γβ1.γ5

s1,s2
6 cw MDelta sw

{{n, 1}, {Δ-, 2}, {π+, 3}} -  hD ϵμ2,α1,β1,γ1 p2

β1 p3

α1 γγ1.γ5

s1,s2
MDelta fπ

{{n, 1}, {Δ-, 2}, {W+, 3}} - e hD Vud ϵμ2,μ3,α1,β1 p2

α1 γβ1.γ5

s1,s2
2 MDelta sw

{{n, 1}, {Δ+, 2}, {π-, 3}}

 hD ϵμ2,α1,β1,γ1 p2

β1 p3

α1 γγ1.γ5

s1,s2
3 MDelta fπ

{{n, 1}, {Δ+, 2}, {W-, 3}}

e hD Vud ϵμ2,μ3,α1,β1 p2

α1 γβ1.γ5

s1,s2
2 3 MDelta sw

{n, 1}, {n, 2}, π0, 3 - ga SlashedP[3].γ5

s1,s2
2 fπ

{{n, 1}, {n, 2}, {Z, 3}} -  e γs1,s2

μ3

4 cw sw

+  e ga γμ3.γ5

s1,s2
4 cw sw

{{n, 1}, {p, 2}, {π-, 3}} ga SlashedP[3].γ5

s1,s2
2 fπ

{{n, 1}, {p, 2}, {W-, 3}}

 e Vud γs1,s2

μ3

2 2 sw

-  e ga Vud γμ3.γ5

s1,s2
2 2 sw

π0, 1, {π+, 2}, {W-, 3}  e Vud p1

μ3

2 sw

-  e Vud p2

μ3

2 sw

π0, 1, {π-, 2}, {W+, 3} -  e Vud p1

μ3

2 sw

+  e Vud p2

μ3

2 sw

{{π+, 1}, {π-, 2}, {Z, 3}}

 e p1

μ3

2 cw sw

-  e sw p1

μ3

cw

-  e p2

μ3

2 cw sw

+  e sw p2

μ3

cw

{{π+, 1}, {W-, 2}, {Z, 3}} e2 Vud ημ2,μ3 fπ

2 cw

{{π-, 1}, {W+, 2}, {Z, 3}} - e2 Vud ημ2,μ3 fπ

2 cw

{p, 1}, Δ0, 2, {π+, 3} -  hD ϵμ2,α1,β1,γ1 p2

β1 p3

α1 γγ1.γ5

s1,s2
3 MDelta fπ

{p, 1}, Δ0, 2, {W+, 3} - e hD Vud ϵμ2,μ3,α1,β1 p2

α1 γβ1.γ5

s1,s2
2 3 MDelta sw

{{p, 1}, {Δ+, 2}, {A, 3}}  2

3 G2 Kappa p2

μ3 p3

μ2 γs1,s2

5 - G1 Kappa ϵμ2,μ3,α1,β1 p2

α1 p3

β1 δs2,s1
6 - G1 Kappa ϵμ2,μ3,α1,γ1 p2

α1 p3

γ1 δs2,s1
6 -  2

3 G2 Kappa γs1,s2

5 ημ2,μ3 p2.p3

{p, 1}, {Δ+, 2}, π0, 3 -

 2
3 hD ϵμ2,α1,β1,γ1 p2

β1 p3

α1 γγ1.γ5

s1,s2
MDelta fπ

{{p, 1}, {Δ+, 2}, {Z, 3}} - e hD ϵμ2,μ3,α1,β1 p2

α1 γβ1.γ5

s1,s2
6 cw MDelta sw

{{p, 1}, {Δ++, 2}, {π-, 3}}

 hD ϵμ2,α1,β1,γ1 p2

β1 p3

α1 γγ1.γ5

s1,s2
MDelta fπ

{{p, 1}, {Δ++, 2}, {W-, 3}}

e hD Vud ϵμ2,μ3,α1,β1 p2

α1 γβ1.γ5

s1,s2
2 MDelta sw

{{p, 1}, {n, 2}, {π+, 3}} ga SlashedP[3].γ5

s1,s2
2 fπ

{{p, 1}, {n, 2}, {W+, 3}}

 e Vud γs1,s2

μ3

2 2 sw

-  e ga Vud γμ3.γ5

s1,s2
2 2 sw

{{p, 1}, {p, 2}, {A, 3}}  e γs1,s2

μ3

{p, 1}, {p, 2}, π0, 3 ga SlashedP[3].γ5

s1,s2
2 fπ

{{p, 1}, {p, 2}, {Z, 3}}

 e γs1,s2

μ3

4 cw sw

-  e sw γs1,s2

μ3

cw

-  e ga γμ3.γ5

s1,s2
4 cw sw

{{A, 1}, {A, 2}, {π+, 3}, {π-, 4}} 2  e2 ημ1,μ2

{A, 1}, π0, 2, {π+, 3}, {W-, 4} -  e2 Vud ημ1,μ4

2 sw

{A, 1}, π0, 2, {π-, 3}, {W+, 4} -  e2 Vud ημ1,μ4

2 sw

{{A, 1}, {π+, 2}, {π-, 3}, {Z, 4}}

 e2 ημ1,μ4
cw sw

- 2  e2 sw ημ1,μ4
cw

Δ0, 1, Δ0, 2, {π+, 3}, {π-, 4} -  AD p3

μ2 γs1,s2

μ1

4 fπ2 +  AD p4

μ2 γs1,s2

μ1

4 fπ2 -  AD p3

μ1 γs1,s2

μ2

4 fπ2 +  AD p4

μ1 γs1,s2

μ2

4 fπ2 -  ημ1,μ2 γ.p3s1,s2

4 fπ2 +  ημ1,μ2 γ.p4s1,s2

4 fπ2 -  γμ1.SlashedP[3].γμ2s1,s2

8 fπ2 -

 AD γμ1.SlashedP[3].γμ2s1,s2

4 fπ2 - 3  AD2 γμ1.SlashedP[3].γμ2s1,s2

8 fπ2 +  γμ1.SlashedP[4].γμ2s1,s2

8 fπ2 +  AD γμ1.SlashedP[4].γμ2s1,s2

4 fπ2 + 3  AD2 γμ1.SlashedP[4].γμ2s1,s2

8 fπ2

Δ0, 1, Δ0, 2, {π+, 3}, {W-, 4} e Vud γs1,s2

μ4 ημ1,μ2
4 sw fπ

+ AD e Vud γs1,s2

μ2 ημ1,μ4
4 sw fπ

+ AD e Vud γs1,s2

μ1 ημ2,μ4
4 sw fπ

+ e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

+ AD e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

+ 3 AD2 e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

Δ0, 1, Δ0, 2, {π-, 3}, {W+, 4} - e Vud γs1,s2

μ4 ημ1,μ2
4 sw fπ

- AD e Vud γs1,s2

μ2 ημ1,μ4
4 sw fπ

- AD e Vud γs1,s2

μ1 ημ2,μ4
4 sw fπ

- e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

- AD e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

- 3 AD2 e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

Δ0, 1, {Δ-, 2}, π0, 3, {π+, 4}  3

2 AD p3

μ2 γs1,s2

μ1

2 fπ2 -

 3
2 AD p4

μ2 γs1,s2

μ1

2 fπ2 +

 3
2 AD p3

μ1 γs1,s2

μ2

2 fπ2 -

 3
2 AD p4

μ1 γs1,s2

μ2

2 fπ2 +

 3
2 ημ1,μ2 γ.p3s1,s2
2 fπ2 -

 3
2 ημ1,μ2 γ.p4s1,s2

2 fπ2 +

 3
2 γμ1.SlashedP[3].γμ2s1,s2
4 fπ2 +

 3
2 AD γμ1.SlashedP[3].γμ2s1,s2
2 fπ2 +

3  3
2 AD2 γμ1.SlashedP[3].γμ2s1,s2
4 fπ2 -

 3
2 γμ1.SlashedP[4].γμ2s1,s2
4 fπ2 -

 3
2 AD γμ1.SlashedP[4].γμ2s1,s2
2 fπ2 -

3  3
2 AD2 γμ1.SlashedP[4].γμ2s1,s2
4 fπ2

Δ0, 1, {Δ-, 2}, π0, 3, {W+, 4} -

3

2 e Vud γs1,s2

μ4 ημ1,μ2
2 sw fπ

-

3

2 AD e Vud γs1,s2

μ2 ημ1,μ4
2 sw fπ

-

3

2 AD e Vud γs1,s2

μ1 ημ2,μ4
2 sw fπ

-

3

2 e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

-

3

2 AD e Vud γμ1.γμ4.γμ2s1,s2
2 sw fπ

-

3 3
2 AD2 e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

Δ0, 1, {Δ-, 2}, {π+, 3}, {Z, 4} e γs1,s2

μ4 ημ1,μ2
6 cw sw fπ

+ cw e γs1,s2

μ4 ημ1,μ2
2 6 sw fπ

+ e sw γs1,s2

μ4 ημ1,μ2
2 6 cw fπ

+ AD e γs1,s2

μ2 ημ1,μ4
6 cw sw fπ

+ AD cw e γs1,s2

μ2 ημ1,μ4
2 6 sw fπ

+ AD e sw γs1,s2

μ2 ημ1,μ4
2 6 cw fπ

+

AD e γs1,s2

μ1 ημ2,μ4
6 cw sw fπ

+ AD cw e γs1,s2

μ1 ημ2,μ4
2 6 sw fπ

+ AD e sw γs1,s2

μ1 ημ2,μ4
2 6 cw fπ

+ e γμ1.γμ4.γμ2s1,s2
2 6 cw sw fπ

+ AD e γμ1.γμ4.γμ2s1,s2
6 cw sw fπ

+

3

2 AD2 e γμ1.γμ4.γμ2s1,s2
2 cw sw fπ

+

cw e γμ1.γμ4.γμ2s1,s2
4 6 sw fπ

+ AD cw e γμ1.γμ4.γμ2s1,s2
2 6 sw fπ

+

3

2 AD2 cw e γμ1.γμ4.γμ2s1,s2
4 sw fπ

+ e sw γμ1.γμ4.γμ2s1,s2
4 6 cw fπ

+ AD e sw γμ1.γμ4.γμ2s1,s2
2 6 cw fπ

+

3

2 AD2 e sw γμ1.γμ4.γμ2s1,s2
4 cw fπ

Δ0, 1, {Δ+, 2}, π0, 3, {π-, 4} -  AD p3

μ2 γs1,s2

μ1

2 fπ2 +  AD p4

μ2 γs1,s2

μ1

2 fπ2 -  AD p3

μ1 γs1,s2

μ2

2 fπ2 +  AD p4

μ1 γs1,s2

μ2

2 fπ2 -  ημ1,μ2 γ.p3s1,s2

2 fπ2 +  ημ1,μ2 γ.p4s1,s2

2 fπ2 -  γμ1.SlashedP[3].γμ2s1,s2

2 2 fπ2 -

 AD γμ1.SlashedP[3].γμ2s1,s2

2 fπ2 - 3  AD2 γμ1.SlashedP[3].γμ2s1,s2

2 2 fπ2 +  γμ1.SlashedP[4].γμ2s1,s2

2 2 fπ2 +  AD γμ1.SlashedP[4].γμ2s1,s2

2 fπ2 + 3  AD2 γμ1.SlashedP[4].γμ2s1,s2

2 2 fπ2

Δ0, 1, {Δ+, 2}, π0, 3, {W-, 4} e Vud γs1,s2

μ4 ημ1,μ2
2 sw fπ

+ AD e Vud γs1,s2

μ2 ημ1,μ4
2 sw fπ

+ AD e Vud γs1,s2

μ1 ημ2,μ4
2 sw fπ

+ e Vud γμ1.γμ4.γμ2s1,s2
2 2 sw fπ

+ AD e Vud γμ1.γμ4.γμ2s1,s2
2 sw fπ

+ 3 AD2 e Vud γμ1.γμ4.γμ2s1,s2
2 2 sw fπ

Δ0, 1, {Δ+, 2}, {π-, 3}, {Z, 4} - 2 e γs1,s2

μ4 ημ1,μ2
3 cw sw fπ

- cw e γs1,s2

μ4 ημ1,μ2
3 2 sw fπ

- e sw γs1,s2

μ4 ημ1,μ2
3 2 cw fπ

- 2 AD e γs1,s2

μ2 ημ1,μ4
3 cw sw fπ

- AD cw e γs1,s2

μ2 ημ1,μ4
3 2 sw fπ

- AD e sw γs1,s2

μ2 ημ1,μ4
3 2 cw fπ

-

2 AD e γs1,s2

μ1 ημ2,μ4
3 cw sw fπ

- AD cw e γs1,s2

μ1 ημ2,μ4
3 2 sw fπ

- AD e sw γs1,s2

μ1 ημ2,μ4
3 2 cw fπ

- e γμ1.γμ4.γμ2s1,s2
3 2 cw sw fπ

- 2 AD e γμ1.γμ4.γμ2s1,s2
3 cw sw fπ

- AD2 e γμ1.γμ4.γμ2s1,s2
2 cw sw fπ

-

cw e γμ1.γμ4.γμ2s1,s2
6 2 sw fπ

- AD cw e γμ1.γμ4.γμ2s1,s2
3 2 sw fπ

- AD2 cw e γμ1.γμ4.γμ2s1,s2
2 2 sw fπ

- e sw γμ1.γμ4.γμ2s1,s2
6 2 cw fπ

- AD e sw γμ1.γμ4.γμ2s1,s2
3 2 cw fπ

- AD2 e sw γμ1.γμ4.γμ2s1,s2
2 2 cw fπ

Δ0, 1, {n, 2}, {π+, 3}, {W-, 4}  e hD Vud ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2

6 MDelta sw fπ

Δ0, 1, {n, 2}, {π-, 3}, {W+, 4} -  e hD Vud ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2
6 MDelta sw fπ

Δ0, 1, {p, 2}, {A, 3}, {π-, 4}  e hD ϵμ1,μ3,α1,β1 p1

α1 γβ1.γ5

s1,s2
3 MDelta fπ

Δ0, 1, {p, 2}, π0, 3, {W-, 4} -  e hD Vud ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2
2 3 MDelta sw fπ

Δ0, 1, {p, 2}, {π-, 3}, {Z, 4}  e hD ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2
2 3 cw MDelta sw fπ

-  e hD sw ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2
3 cw MDelta fπ

{Δ-, 1}, Δ0, 2, π0, 3, {π-, 4} -

 3
2 AD p3

μ2 γs1,s2

μ1

2 fπ2 +

 3
2 AD p4

μ2 γs1,s2

μ1

2 fπ2 -

 3
2 AD p3

μ1 γs1,s2

μ2

2 fπ2 +

 3
2 AD p4

μ1 γs1,s2

μ2

2 fπ2 -

 3
2 ημ1,μ2 γ.p3s1,s2

2 fπ2 +

 3
2 ημ1,μ2 γ.p4s1,s2
2 fπ2 -

 3
2 γμ1.SlashedP[3].γμ2s1,s2
4 fπ2 -

 3
2 AD γμ1.SlashedP[3].γμ2s1,s2
2 fπ2 -

3  3
2 AD2 γμ1.SlashedP[3].γμ2s1,s2
4 fπ2 +

 3
2 γμ1.SlashedP[4].γμ2s1,s2
4 fπ2 +

 3
2 AD γμ1.SlashedP[4].γμ2s1,s2
2 fπ2 +

3  3
2 AD2 γμ1.SlashedP[4].γμ2s1,s2
4 fπ2

{Δ-, 1}, Δ0, 2, π0, 3, {W-, 4}

3

2 e Vud γs1,s2

μ4 ημ1,μ2
2 sw fπ

+

3

2 AD e Vud γs1,s2

μ2 ημ1,μ4
2 sw fπ

+

3

2 AD e Vud γs1,s2

μ1 ημ2,μ4
2 sw fπ

+

3

2 e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

+

3

2 AD e Vud γμ1.γμ4.γμ2s1,s2
2 sw fπ

+

3 3
2 AD2 e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

{Δ-, 1}, Δ0, 2, {π-, 3}, {Z, 4} - e γs1,s2

μ4 ημ1,μ2
6 cw sw fπ

- cw e γs1,s2

μ4 ημ1,μ2
2 6 sw fπ

- e sw γs1,s2

μ4 ημ1,μ2
2 6 cw fπ

- AD e γs1,s2

μ2 ημ1,μ4
6 cw sw fπ

- AD cw e γs1,s2

μ2 ημ1,μ4
2 6 sw fπ

- AD e sw γs1,s2

μ2 ημ1,μ4
2 6 cw fπ

-

AD e γs1,s2

μ1 ημ2,μ4
6 cw sw fπ

- AD cw e γs1,s2

μ1 ημ2,μ4
2 6 sw fπ

- AD e sw γs1,s2

μ1 ημ2,μ4
2 6 cw fπ

- e γμ1.γμ4.γμ2s1,s2
2 6 cw sw fπ

- AD e γμ1.γμ4.γμ2s1,s2
6 cw sw fπ

-

3

2 AD2 e γμ1.γμ4.γμ2s1,s2
2 cw sw fπ

-

cw e γμ1.γμ4.γμ2s1,s2
4 6 sw fπ

- AD cw e γμ1.γμ4.γμ2s1,s2
2 6 sw fπ

-

3

2 AD2 cw e γμ1.γμ4.γμ2s1,s2
4 sw fπ

- e sw γμ1.γμ4.γμ2s1,s2
4 6 cw fπ

- AD e sw γμ1.γμ4.γμ2s1,s2
2 6 cw fπ

-

3

2 AD2 e sw γμ1.γμ4.γμ2s1,s2
4 cw fπ

{{Δ-, 1}, {Δ-, 2}, {π+, 3}, {π-, 4}} - 3  AD p3

μ2 γs1,s2

μ1

4 fπ2 + 3  AD p4

μ2 γs1,s2

μ1

4 fπ2 - 3  AD p3

μ1 γs1,s2

μ2

4 fπ2 + 3  AD p4

μ1 γs1,s2

μ2

4 fπ2 - 3  ημ1,μ2 γ.p3s1,s2

4 fπ2 + 3  ημ1,μ2 γ.p4s1,s2

4 fπ2 - 3  γμ1.SlashedP[3].γμ2s1,s2

8 fπ2 -

3  AD γμ1.SlashedP[3].γμ2s1,s2

4 fπ2 - 9  AD2 γμ1.SlashedP[3].γμ2s1,s2

8 fπ2 + 3  γμ1.SlashedP[4].γμ2s1,s2

8 fπ2 + 3  AD γμ1.SlashedP[4].γμ2s1,s2

4 fπ2 + 9  AD2 γμ1.SlashedP[4].γμ2s1,s2

8 fπ2

{{Δ-, 1}, {Δ-, 2}, {π+, 3}, {W-, 4}}

3 e Vud γs1,s2

μ4 ημ1,μ2
4 sw fπ

+ 3 AD e Vud γs1,s2

μ2 ημ1,μ4
4 sw fπ

+ 3 AD e Vud γs1,s2

μ1 ημ2,μ4
4 sw fπ

+ 3 e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

+ 3 AD e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

+ 9 AD2 e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

{{Δ-, 1}, {Δ-, 2}, {π-, 3}, {W+, 4}} - 3 e Vud γs1,s2

μ4 ημ1,μ2
4 sw fπ

- 3 AD e Vud γs1,s2

μ2 ημ1,μ4
4 sw fπ

- 3 AD e Vud γs1,s2

μ1 ημ2,μ4
4 sw fπ

- 3 e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

- 3 AD e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

- 9 AD2 e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

{{Δ-, 1}, {n, 2}, {A, 3}, {π-, 4}}

 e hD ϵμ1,μ3,α1,β1 p1

α1 γβ1.γ5

s1,s2
MDelta fπ

{Δ-, 1}, {n, 2}, π0, 3, {W-, 4} -  e hD Vud ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2
2 MDelta sw fπ

{{Δ-, 1}, {n, 2}, {π-, 3}, {Z, 4}}

 e hD ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2
2 cw MDelta sw fπ

-  e hD sw ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2
cw MDelta fπ

Δ+, 1, Δ0, 2, π0, 3, {π+, 4}  AD p3

μ2 γs1,s2

μ1

2 fπ2 -  AD p4

μ2 γs1,s2

μ1

2 fπ2 +  AD p3

μ1 γs1,s2

μ2

2 fπ2 -  AD p4

μ1 γs1,s2

μ2

2 fπ2 +  ημ1,μ2 γ.p3s1,s2

2 fπ2 -  ημ1,μ2 γ.p4s1,s2

2 fπ2 +  γμ1.SlashedP[3].γμ2s1,s2

2 2 fπ2 +

 AD γμ1.SlashedP[3].γμ2s1,s2

2 fπ2 + 3  AD2 γμ1.SlashedP[3].γμ2s1,s2

2 2 fπ2 -  γμ1.SlashedP[4].γμ2s1,s2

2 2 fπ2 -  AD γμ1.SlashedP[4].γμ2s1,s2

2 fπ2 - 3  AD2 γμ1.SlashedP[4].γμ2s1,s2

2 2 fπ2

Δ+, 1, Δ0, 2, π0, 3, {W+, 4} - e Vud γs1,s2

μ4 ημ1,μ2
2 sw fπ

- AD e Vud γs1,s2

μ2 ημ1,μ4
2 sw fπ

- AD e Vud γs1,s2

μ1 ημ2,μ4
2 sw fπ

- e Vud γμ1.γμ4.γμ2s1,s2
2 2 sw fπ

- AD e Vud γμ1.γμ4.γμ2s1,s2
2 sw fπ

- 3 AD2 e Vud γμ1.γμ4.γμ2s1,s2
2 2 sw fπ

Δ+, 1, Δ0, 2, {π+, 3}, {Z, 4} 2 e γs1,s2

μ4 ημ1,μ2
3 cw sw fπ

+ cw e γs1,s2

μ4 ημ1,μ2
3 2 sw fπ

+ e sw γs1,s2

μ4 ημ1,μ2
3 2 cw fπ

+ 2 AD e γs1,s2

μ2 ημ1,μ4
3 cw sw fπ

+ AD cw e γs1,s2

μ2 ημ1,μ4
3 2 sw fπ

+ AD e sw γs1,s2

μ2 ημ1,μ4
3 2 cw fπ

+

2 AD e γs1,s2

μ1 ημ2,μ4
3 cw sw fπ

+ AD cw e γs1,s2

μ1 ημ2,μ4
3 2 sw fπ

+ AD e sw γs1,s2

μ1 ημ2,μ4
3 2 cw fπ

+ e γμ1.γμ4.γμ2s1,s2
3 2 cw sw fπ

+ 2 AD e γμ1.γμ4.γμ2s1,s2
3 cw sw fπ

+ AD2 e γμ1.γμ4.γμ2s1,s2
2 cw sw fπ

+

cw e γμ1.γμ4.γμ2s1,s2
6 2 sw fπ

+ AD cw e γμ1.γμ4.γμ2s1,s2
3 2 sw fπ

+ AD2 cw e γμ1.γμ4.γμ2s1,s2
2 2 sw fπ

+ e sw γμ1.γμ4.γμ2s1,s2
6 2 cw fπ

+ AD e sw γμ1.γμ4.γμ2s1,s2
3 2 cw fπ

+ AD2 e sw γμ1.γμ4.γμ2s1,s2
2 2 cw fπ

Δ+, 1, {Δ+, 2}, {π+, 3}, {π-, 4}  AD p3

μ2 γs1,s2

μ1

4 fπ2 -  AD p4

μ2 γs1,s2

μ1

4 fπ2 +  AD p3

μ1 γs1,s2

μ2

4 fπ2 -  AD p4

μ1 γs1,s2

μ2

4 fπ2 +  ημ1,μ2 γ.p3s1,s2

4 fπ2 -  ημ1,μ2 γ.p4s1,s2

4 fπ2 +  γμ1.SlashedP[3].γμ2s1,s2

8 fπ2 +

 AD γμ1.SlashedP[3].γμ2s1,s2

4 fπ2 + 3  AD2 γμ1.SlashedP[3].γμ2s1,s2

8 fπ2 -  γμ1.SlashedP[4].γμ2s1,s2

8 fπ2 -  AD γμ1.SlashedP[4].γμ2s1,s2

4 fπ2 - 3  AD2 γμ1.SlashedP[4].γμ2s1,s2

8 fπ2

Δ+, 1, {Δ+, 2}, {π+, 3}, {W-, 4} - e Vud γs1,s2

μ4 ημ1,μ2
4 sw fπ

- AD e Vud γs1,s2

μ2 ημ1,μ4
4 sw fπ

- AD e Vud γs1,s2

μ1 ημ2,μ4
4 sw fπ

- e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

- AD e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

- 3 AD2 e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

Δ+, 1, {Δ+, 2}, {π-, 3}, {W+, 4} e Vud γs1,s2

μ4 ημ1,μ2
4 sw fπ

+ AD e Vud γs1,s2

μ2 ημ1,μ4
4 sw fπ

+ AD e Vud γs1,s2

μ1 ημ2,μ4
4 sw fπ

+ e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

+ AD e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

+ 3 AD2 e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

Δ+, 1, {Δ++, 2}, π0, 3, {π-, 4} -

 3
2 AD p3

μ2 γs1,s2

μ1

2 fπ2 +

 3
2 AD p4

μ2 γs1,s2

μ1

2 fπ2 -

 3
2 AD p3

μ1 γs1,s2

μ2

2 fπ2 +

 3
2 AD p4

μ1 γs1,s2

μ2

2 fπ2 -

 3
2 ημ1,μ2 γ.p3s1,s2

2 fπ2 +

 3
2 ημ1,μ2 γ.p4s1,s2
2 fπ2 -

 3
2 γμ1.SlashedP[3].γμ2s1,s2
4 fπ2 -

 3
2 AD γμ1.SlashedP[3].γμ2s1,s2
2 fπ2 -

3  3
2 AD2 γμ1.SlashedP[3].γμ2s1,s2
4 fπ2 +

 3
2 γμ1.SlashedP[4].γμ2s1,s2
4 fπ2 +

 3
2 AD γμ1.SlashedP[4].γμ2s1,s2
2 fπ2 +

3  3
2 AD2 γμ1.SlashedP[4].γμ2s1,s2
4 fπ2

Δ+, 1, {Δ++, 2}, π0, 3, {W-, 4}

3

2 e Vud γs1,s2

μ4 ημ1,μ2
2 sw fπ

+

3

2 AD e Vud γs1,s2

μ2 ημ1,μ4
2 sw fπ

+

3

2 AD e Vud γs1,s2

μ1 ημ2,μ4
2 sw fπ

+

3

2 e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

+

3

2 AD e Vud γμ1.γμ4.γμ2s1,s2
2 sw fπ

+

3 3
2 AD2 e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

Δ+, 1, {Δ++, 2}, {π-, 3}, {Z, 4} - e γs1,s2

μ4 ημ1,μ2
6 cw sw fπ

- cw e γs1,s2

μ4 ημ1,μ2
2 6 sw fπ

- e sw γs1,s2

μ4 ημ1,μ2
2 6 cw fπ

- AD e γs1,s2

μ2 ημ1,μ4
6 cw sw fπ

- AD cw e γs1,s2

μ2 ημ1,μ4
2 6 sw fπ

- AD e sw γs1,s2

μ2 ημ1,μ4
2 6 cw fπ

-

AD e γs1,s2

μ1 ημ2,μ4
6 cw sw fπ

- AD cw e γs1,s2

μ1 ημ2,μ4
2 6 sw fπ

- AD e sw γs1,s2

μ1 ημ2,μ4
2 6 cw fπ

- e γμ1.γμ4.γμ2s1,s2
2 6 cw sw fπ

- AD e γμ1.γμ4.γμ2s1,s2
6 cw sw fπ

-

3

2 AD2 e γμ1.γμ4.γμ2s1,s2
2 cw sw fπ

-

cw e γμ1.γμ4.γμ2s1,s2
4 6 sw fπ

- AD cw e γμ1.γμ4.γμ2s1,s2
2 6 sw fπ

-

3

2 AD2 cw e γμ1.γμ4.γμ2s1,s2
4 sw fπ

- e sw γμ1.γμ4.γμ2s1,s2
4 6 cw fπ

- AD e sw γμ1.γμ4.γμ2s1,s2
2 6 cw fπ

-

3

2 AD2 e sw γμ1.γμ4.γμ2s1,s2
4 cw fπ

Δ+, 1, {n, 2}, {A, 3}, {π+, 4}  e hD ϵμ1,μ3,α1,β1 p1

α1 γβ1.γ5

s1,s2
3 MDelta fπ

Δ+, 1, {n, 2}, π0, 3, {W+, 4} -  e hD Vud ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2
2 3 MDelta sw fπ

Δ+, 1, {n, 2}, {π+, 3}, {Z, 4}  e hD ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2
2 3 cw MDelta sw fπ

-  e hD sw ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2
3 cw MDelta fπ

Δ+, 1, {p, 2}, {π+, 3}, {W-, 4}  e hD Vud ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2
6 MDelta sw fπ

Δ+, 1, {p, 2}, {π-, 3}, {W+, 4} -  e hD Vud ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2
6 MDelta sw fπ

Δ++, 1, {Δ+, 2}, π0, 3, {π+, 4}

 3
2 AD p3

μ2 γs1,s2

μ1

2 fπ2 -

 3
2 AD p4

μ2 γs1,s2

μ1

2 fπ2 +

 3
2 AD p3

μ1 γs1,s2

μ2

2 fπ2 -

 3
2 AD p4

μ1 γs1,s2

μ2

2 fπ2 +

 3
2 ημ1,μ2 γ.p3s1,s2
2 fπ2 -

 3
2 ημ1,μ2 γ.p4s1,s2

2 fπ2 +

 3
2 γμ1.SlashedP[3].γμ2s1,s2
4 fπ2 +

 3
2 AD γμ1.SlashedP[3].γμ2s1,s2
2 fπ2 +

3  3
2 AD2 γμ1.SlashedP[3].γμ2s1,s2
4 fπ2 -

 3
2 γμ1.SlashedP[4].γμ2s1,s2
4 fπ2 -

 3
2 AD γμ1.SlashedP[4].γμ2s1,s2
2 fπ2 -

3  3
2 AD2 γμ1.SlashedP[4].γμ2s1,s2
4 fπ2

Δ++, 1, {Δ+, 2}, π0, 3, {W+, 4} -

3

2 e Vud γs1,s2

μ4 ημ1,μ2
2 sw fπ

-

3

2 AD e Vud γs1,s2

μ2 ημ1,μ4
2 sw fπ

-

3

2 AD e Vud γs1,s2

μ1 ημ2,μ4
2 sw fπ

-

3

2 e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

-

3

2 AD e Vud γμ1.γμ4.γμ2s1,s2
2 sw fπ

-

3 3
2 AD2 e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

Δ++, 1, {Δ+, 2}, {π+, 3}, {Z, 4} e γs1,s2

μ4 ημ1,μ2
6 cw sw fπ

+ cw e γs1,s2

μ4 ημ1,μ2
2 6 sw fπ

+ e sw γs1,s2

μ4 ημ1,μ2
2 6 cw fπ

+ AD e γs1,s2

μ2 ημ1,μ4
6 cw sw fπ

+ AD cw e γs1,s2

μ2 ημ1,μ4
2 6 sw fπ

+ AD e sw γs1,s2

μ2 ημ1,μ4
2 6 cw fπ

+

AD e γs1,s2

μ1 ημ2,μ4
6 cw sw fπ

+ AD cw e γs1,s2

μ1 ημ2,μ4
2 6 sw fπ

+ AD e sw γs1,s2

μ1 ημ2,μ4
2 6 cw fπ

+ e γμ1.γμ4.γμ2s1,s2
2 6 cw sw fπ

+ AD e γμ1.γμ4.γμ2s1,s2
6 cw sw fπ

+

3

2 AD2 e γμ1.γμ4.γμ2s1,s2
2 cw sw fπ

+

cw e γμ1.γμ4.γμ2s1,s2
4 6 sw fπ

+ AD cw e γμ1.γμ4.γμ2s1,s2
2 6 sw fπ

+

3

2 AD2 cw e γμ1.γμ4.γμ2s1,s2
4 sw fπ

+ e sw γμ1.γμ4.γμ2s1,s2
4 6 cw fπ

+ AD e sw γμ1.γμ4.γμ2s1,s2
2 6 cw fπ

+

3

2 AD2 e sw γμ1.γμ4.γμ2s1,s2
4 cw fπ

Δ++, 1, {Δ++, 2}, {π+, 3}, {π-, 4} 3  AD p3

μ2 γs1,s2

μ1

4 fπ2 - 3  AD p4

μ2 γs1,s2

μ1

4 fπ2 + 3  AD p3

μ1 γs1,s2

μ2

4 fπ2 - 3  AD p4

μ1 γs1,s2

μ2

4 fπ2 + 3  ημ1,μ2 γ.p3s1,s2

4 fπ2 - 3  ημ1,μ2 γ.p4s1,s2

4 fπ2 + 3  γμ1.SlashedP[3].γμ2s1,s2

8 fπ2 +

3  AD γμ1.SlashedP[3].γμ2s1,s2

4 fπ2 + 9  AD2 γμ1.SlashedP[3].γμ2s1,s2

8 fπ2 - 3  γμ1.SlashedP[4].γμ2s1,s2

8 fπ2 - 3  AD γμ1.SlashedP[4].γμ2s1,s2

4 fπ2 - 9  AD2 γμ1.SlashedP[4].γμ2s1,s2

8 fπ2

Δ++, 1, {Δ++, 2}, {π+, 3}, {W-, 4} - 3 e Vud γs1,s2

μ4 ημ1,μ2
4 sw fπ

- 3 AD e Vud γs1,s2

μ2 ημ1,μ4
4 sw fπ

- 3 AD e Vud γs1,s2

μ1 ημ2,μ4
4 sw fπ

- 3 e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

- 3 AD e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

- 9 AD2 e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

Δ++, 1, {Δ++, 2}, {π-, 3}, {W+, 4} 3 e Vud γs1,s2

μ4 ημ1,μ2
4 sw fπ

+ 3 AD e Vud γs1,s2

μ2 ημ1,μ4
4 sw fπ

+ 3 AD e Vud γs1,s2

μ1 ημ2,μ4
4 sw fπ

+ 3 e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

+ 3 AD e Vud γμ1.γμ4.γμ2s1,s2
4 sw fπ

+ 9 AD2 e Vud γμ1.γμ4.γμ2s1,s2
8 sw fπ

Δ++, 1, {p, 2}, {A, 3}, {π+, 4}  e hD ϵμ1,μ3,α1,β1 p1

α1 γβ1.γ5

s1,s2
MDelta fπ

Δ++, 1, {p, 2}, π0, 3, {W+, 4} -  e hD Vud ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2
2 MDelta sw fπ

Δ++, 1, {p, 2}, {π+, 3}, {Z, 4}  e hD ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2
2 cw MDelta sw fπ

-  e hD sw ϵμ1,μ4,α1,β1 p1

α1 γβ1.γ5

s1,s2
cw MDelta fπ

{n, 1}, Δ0, 2, {π+, 3}, {W-, 4}  e hD Vud ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2
6 MDelta sw fπ

{n, 1}, Δ0, 2, {π-, 3}, {W+, 4} -  e hD Vud ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2
6 MDelta sw fπ

{{n, 1}, {Δ-, 2}, {A, 3}, {π+, 4}} -  e hD ϵμ2,μ3,α1,β1 p2

α1 γβ1.γ5

s1,s2
MDelta fπ

{n, 1}, {Δ-, 2}, π0, 3, {W+, 4}  e hD Vud ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2
2 MDelta sw fπ

{{n, 1}, {Δ-, 2}, {π+, 3}, {Z, 4}} -  e hD ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2
2 cw MDelta sw fπ

+  e hD sw ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2
cw MDelta fπ

{{n, 1}, {Δ+, 2}, {A, 3}, {π-, 4}} -  e hD ϵμ2,μ3,α1,β1 p2

α1 γβ1.γ5

s1,s2
3 MDelta fπ

{n, 1}, {Δ+, 2}, π0, 3, {W-, 4}  e hD Vud ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2
2 3 MDelta sw fπ

{{n, 1}, {Δ+, 2}, {π-, 3}, {Z, 4}} -  e hD ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2
2 3 cw MDelta sw fπ

+  e hD sw ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2
3 cw MDelta fπ

{{n, 1}, {n, 2}, {π+, 3}, {π-, 4}}

 γ.p3s1,s2

4 fπ2 -  γ.p4s1,s2

4 fπ2

{{n, 1}, {n, 2}, {π+, 3}, {W-, 4}} - e Vud γs1,s2

μ4

4 sw fπ

+ e ga Vud γμ4.γ5

s1,s2
4 sw fπ

{{n, 1}, {n, 2}, {π-, 3}, {W+, 4}} e Vud γs1,s2

μ4

4 sw fπ

- e ga Vud γμ4.γ5

s1,s2

4 sw fπ

{{n, 1}, {p, 2}, {A, 3}, {π-, 4}} - e ga γμ3.γ5

s1,s2
2 fπ

{n, 1}, {p, 2}, π0, 3, {π-, 4}  γ.p3s1,s2

2 2 fπ2 -  γ.p4s1,s2

2 2 fπ2

{n, 1}, {p, 2}, π0, 3, {W-, 4} - e Vud γs1,s2

μ4

2 2 sw fπ

+ e ga Vud γμ4.γ5

s1,s2
2 2 sw fπ

{{n, 1}, {p, 2}, {π-, 3}, {Z, 4}} e γs1,s2

μ4

2 2 cw sw fπ

- e ga γμ4.γ5

s1,s2
2 2 cw sw fπ

+ e ga sw γμ4.γ5

s1,s2
2 cw fπ

π0, 1, π0, 2, π0, 3, π0, 4 - 2  p1.p2

fπ2 - 2  p1.p3

fπ2 - 2  p1.p4

fπ2 - 2  p2.p3

fπ2 - 2  p2.p4

fπ2 - 2  p3.p4

fπ2

π0, 1, π0, 2, π0, 3, {Z, 4} - e p1

μ4

2 cw sw fπ

- e p2

μ4

2 cw sw fπ

- e p3

μ4

2 cw sw fπ

π0, 1, π0, 2, {π+, 3}, {π-, 4} -  p1.p3

fπ2 -  p1.p4

fπ2 -  p2.p3

fπ2 -  p2.p4

fπ2

π0, 1, π0, 2, {π+, 3}, {W-, 4} - e Vud p1

μ4

2 sw fπ

- e Vud p2

μ4

2 sw fπ

+ e Vud p3

μ4

2 sw fπ

π0, 1, π0, 2, {π-, 3}, {W+, 4} - e Vud p1

μ4

2 sw fπ

- e Vud p2

μ4

2 sw fπ

+ e Vud p3

μ4

2 sw fπ

π0, 1, {π+, 2}, {π-, 3}, {Z, 4} e p1

μ4

2 cw sw fπ

- e p2

μ4

2 cw sw fπ

- e p3

μ4

2 cw sw fπ

π0, 1, {π+, 2}, {W-, 3}, {Z, 4}  e2 Vud ημ3,μ4

2 cw

π0, 1, {π-, 2}, {W+, 3}, {Z, 4}  e2 Vud ημ3,μ4

2 cw

{{π+, 1}, {π+, 2}, {π-, 3}, {π-, 4}} - 2  p1.p2
fπ2 -  p1.p3

fπ2 -  p1.p4

fπ2 -  p2.p3

fπ2 -  p2.p4

fπ2 - 2  p3.p4

fπ2

{{π+, 1}, {π+, 2}, {π-, 3}, {W-, 4}} - e Vud p3

μ4

sw fπ

{{π+, 1}, {π-, 2}, {π-, 3}, {W+, 4}} - e Vud p1

μ4

sw fπ

{{π+, 1}, {π-, 2}, {Z, 3}, {Z, 4}} -2  e2 ημ3,μ4

{p, 1}, Δ0, 2, {A, 3}, {π+, 4} -  e hD ϵμ2,μ3,α1,β1 p2

α1 γβ1.γ5

s1,s2
3 MDelta fπ

{p, 1}, Δ0, 2, π0, 3, {W+, 4}  e hD Vud ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2
2 3 MDelta sw fπ

{p, 1}, Δ0, 2, {π+, 3}, {Z, 4} -  e hD ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2
2 3 cw MDelta sw fπ

+  e hD sw ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2
3 cw MDelta fπ

{{p, 1}, {Δ+, 2}, {π+, 3}, {W-, 4}}

 e hD Vud ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2

6 MDelta sw fπ

{{p, 1}, {Δ+, 2}, {π-, 3}, {W+, 4}} -  e hD Vud ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2
6 MDelta sw fπ

{{p, 1}, {Δ++, 2}, {A, 3}, {π-, 4}} -  e hD ϵμ2,μ3,α1,β1 p2

α1 γβ1.γ5

s1,s2
MDelta fπ

{p, 1}, {Δ++, 2}, π0, 3, {W-, 4}  e hD Vud ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2
2 MDelta sw fπ

{{p, 1}, {Δ++, 2}, {π-, 3}, {Z, 4}} -  e hD ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2
2 cw MDelta sw fπ

+  e hD sw ϵμ2,μ4,α1,β1 p2

α1 γβ1.γ5

s1,s2
cw MDelta fπ

{{p, 1}, {n, 2}, {A, 3}, {π+, 4}} e ga γμ3.γ5

s1,s2
2 fπ

{p, 1}, {n, 2}, π0, 3, {π+, 4} -  γ.p3s1,s2

2 2 fπ2 +  γ.p4s1,s2

2 2 fπ2

{p, 1}, {n, 2}, π0, 3, {W+, 4} e Vud γs1,s2

μ4

2 2 sw fπ

- e ga Vud γμ4.γ5

s1,s2
2 2 sw fπ

{{p, 1}, {n, 2}, {π+, 3}, {Z, 4}} - e γs1,s2

μ4

2 2 cw sw fπ

+ e ga γμ4.γ5

s1,s2

2 2 cw sw fπ

- e ga sw γμ4.γ5

s1,s2

2 cw fπ

{{p, 1}, {p, 2}, {π+, 3}, {π-, 4}} -  γ.p3s1,s2

4 fπ2 +  γ.p4s1,s2

4 fπ2

{{p, 1}, {p, 2}, {π+, 3}, {W-, 4}} e Vud γs1,s2

μ4

4 sw fπ

- e ga Vud γμ4.γ5

s1,s2
4 sw fπ

{{p, 1}, {p, 2}, {π-, 3}, {W+, 4}} - e Vud γs1,s2

μ4

4 sw fπ

+ e ga Vud γμ4.γ5

s1,s2
4 sw fπ